

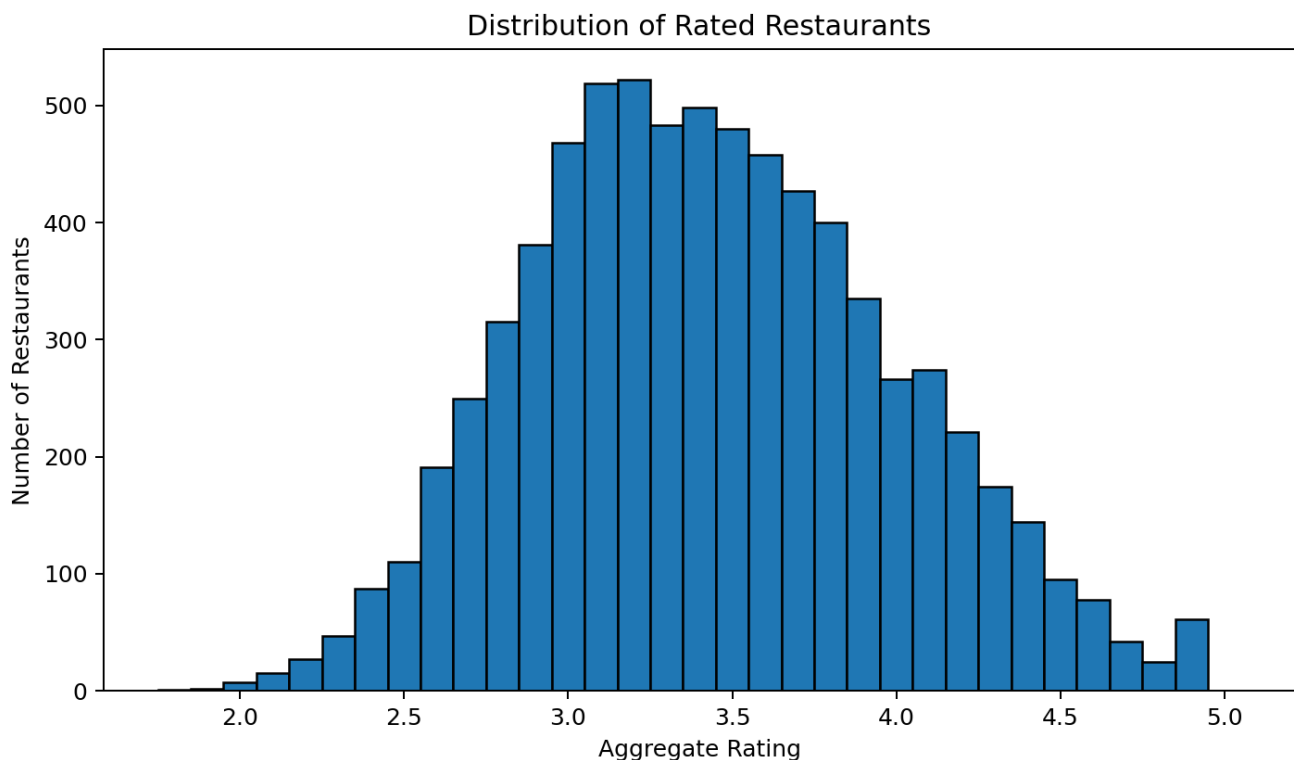
Cognifyz Technologies — Machine Learning Internship

Task 1: Predict Restaurant Ratings

Objective: Build a regression model that predicts a restaurant's aggregate rating from available restaurant, location, service, cost, cuisine and voting features.

1. Dataset & Data Quality

The supplied dataset contains 9,551 restaurant records and 21 columns. There are 9 missing Cuisine values. 2,148 records have an Aggregate rating of 0 and are explicitly labelled 'Not rated', so these records were excluded from regression training rather than treating absence of a rating as a numerical quality score. The remaining rated sample contains 7,403 restaurants.



2. Feature Engineering & Leakage Control

To keep the model defensible, Restaurant ID, Restaurant Name, Address, Locality Verbose, Rating color and Rating text were not used as predictors. In particular, Rating color/text are derived from the target rating and would create target leakage. Categorical variables were handled natively by CatBoost. Missing Cuisine values were assigned to an 'Unknown' category. The data was split 80/20 with a fixed random seed for reproducibility.

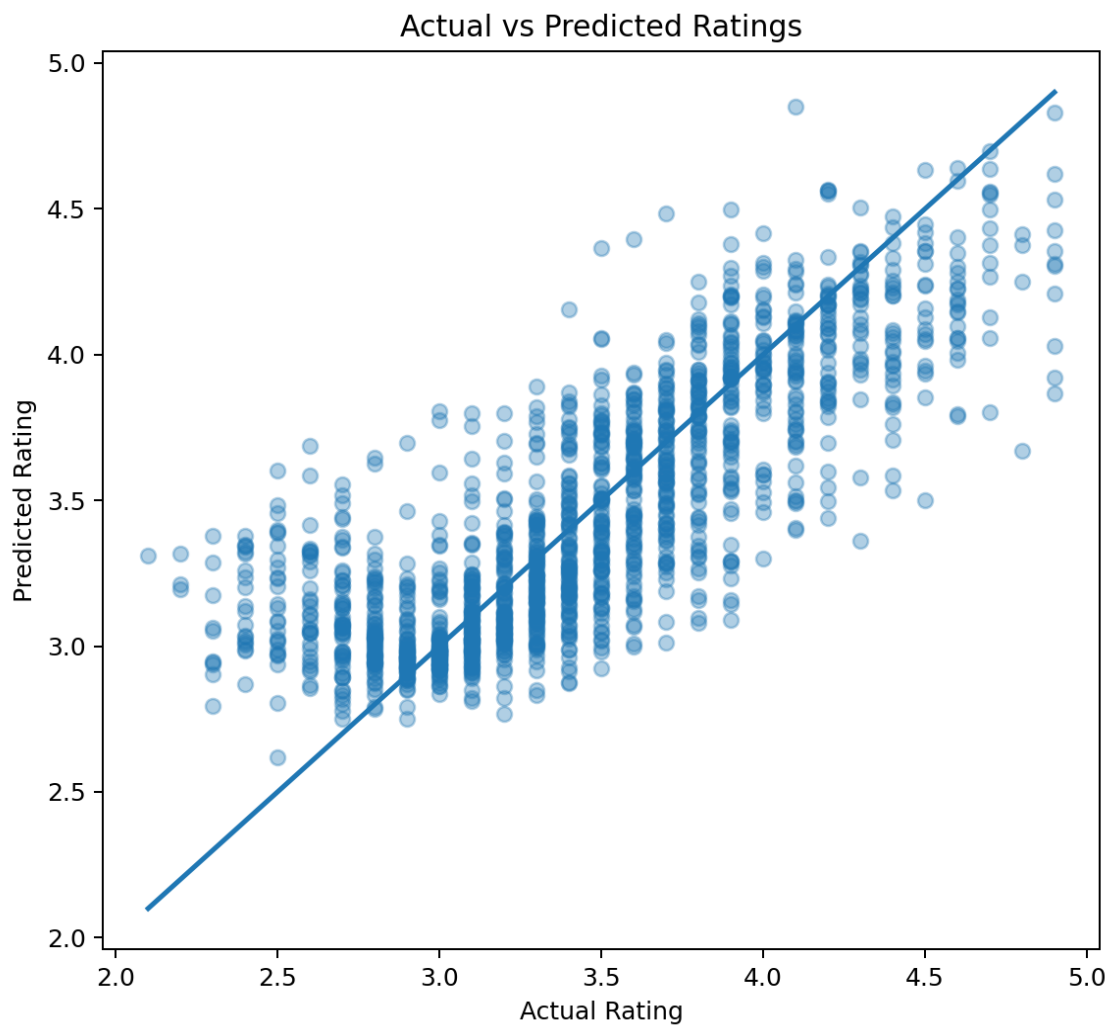
3. Model

A CatBoostRegressor was selected because the dataset contains several high-cardinality categorical features (especially City, Locality and Cuisines). The model was trained with early stopping on the held-out test set and optimized for RMSE.

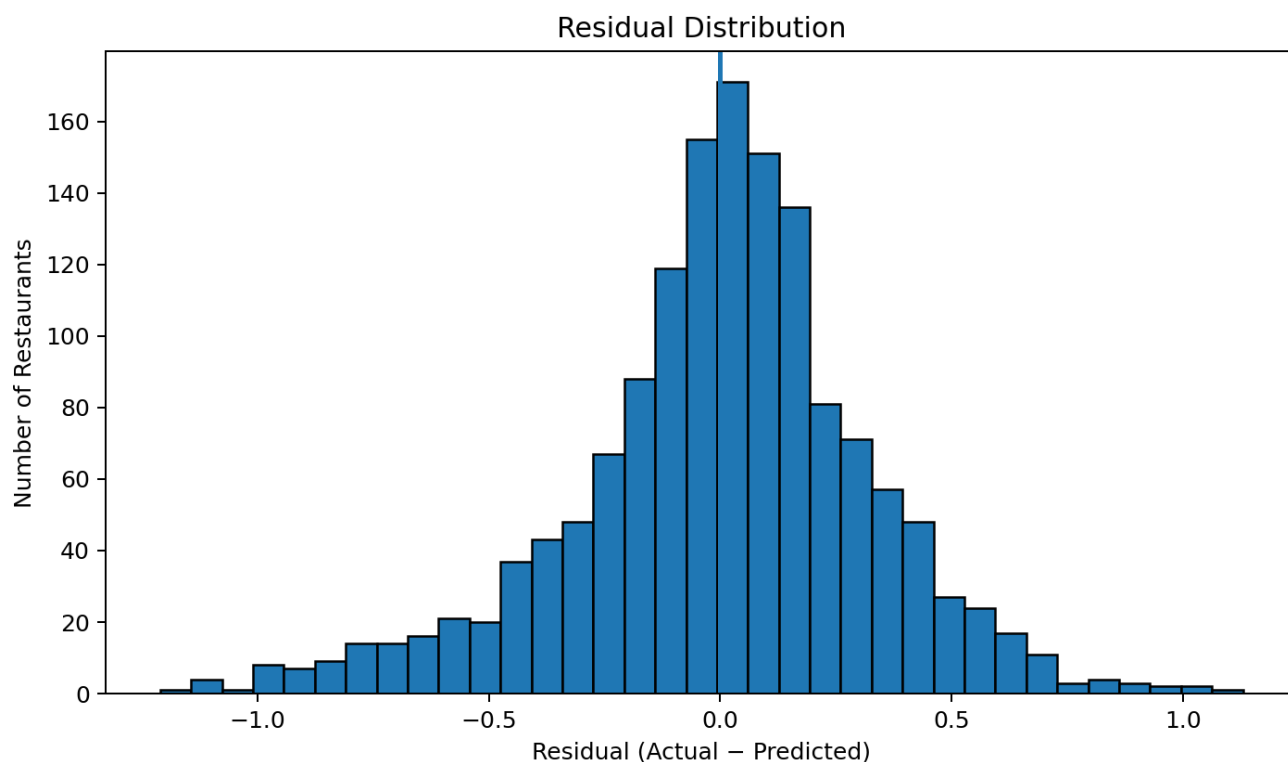
4. Evaluation Results

Metric	Result
RMSE	0.325
MAE	0.241
R ²	0.658

Predictions within ± 0.30 rating	69.7%
Predictions within ± 0.50 rating	87.5%

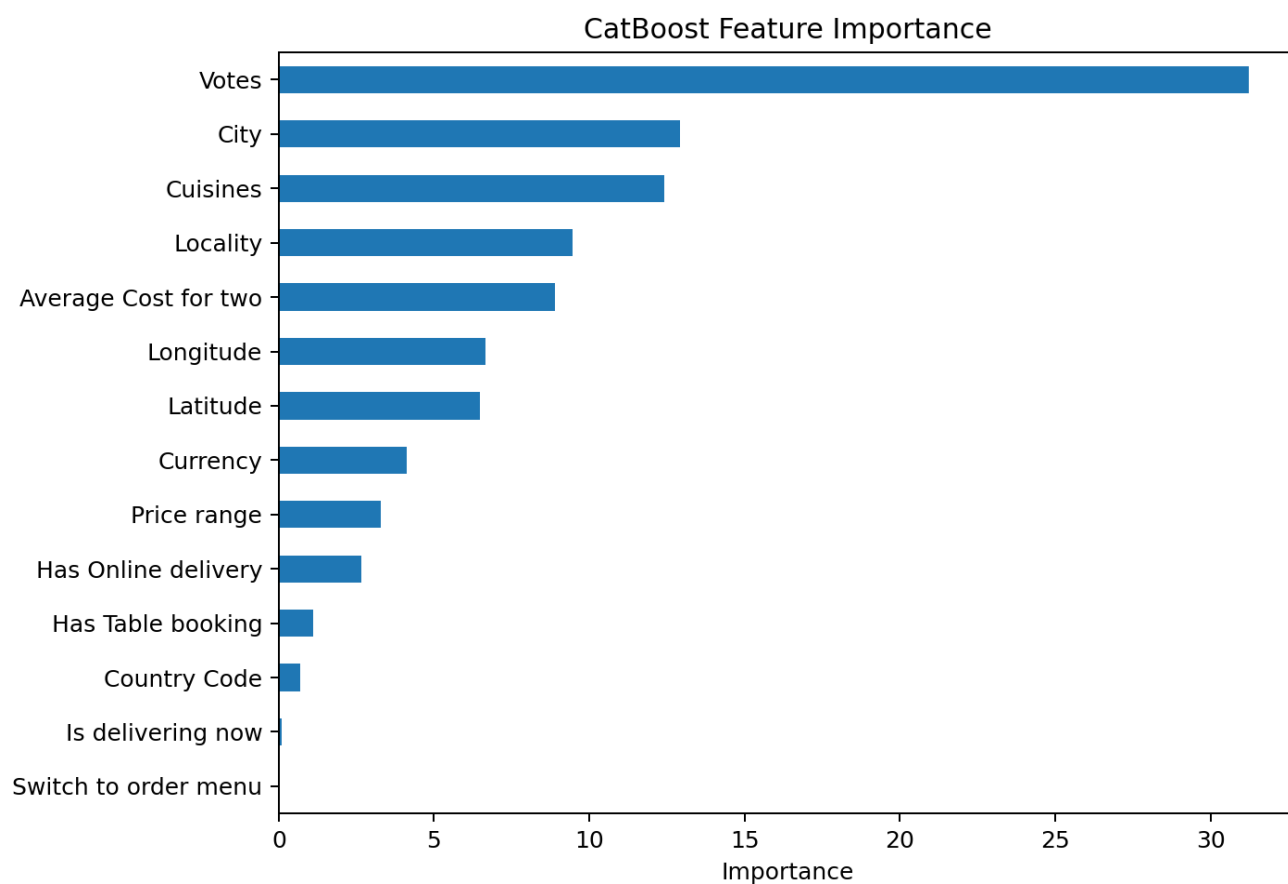


5. Error Analysis



The parity plot shows a strong positive relationship between actual and predicted ratings, while the residual distribution is centered around zero. The model is less precise at the extreme ends of the rating range, which is a useful limitation to mention rather than hiding it.

6. Feature Importance



Top five model features: Votes (31.2), City (12.9), Cuisines (12.4), Locality (9.5), Average Cost for two (8.9). Votes is the strongest predictor in this dataset, followed by geographic/cuisine and cost-related features. Feature importance indicates predictive contribution; it should not be interpreted as causal evidence.

7. Engineering Takeaways

- Treat 'Not rated' as a separate data state rather than a zero-quality rating.
- Use leakage audits before training: target-derived columns can make metrics look artificially strong.
- Native categorical handling is a good fit for high-cardinality restaurant metadata.
- The final model substantially improves over a mean-rating baseline (baseline RMSE ≈ 0.556).
- Future improvement: add temporal/review-volume features if they are legitimately available at prediction time, and evaluate with cross-validation.

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